

3-MONTH AI/ML LEARNING ROADMAP

Math + Python + Core ML + Deep Learning - Build Every Week

by SYNTAX ERROR | syntaxerrorr.com | [@code.abhii07](https://twitter.com/code.abhii07)

3	12	4-5 hrs	1
Months	Weeks	Daily	Project/Week

DAILY STUDY PLAN (4-5 hours)

2 hrs	2 hrs	1 hr
Theory + Study	Coding Practice	Project Build

GOLDEN RULES - FOLLOW THIS

X	No tutorial hopping — pick ONE resource per topic and finish it
X	Don't skip the Math — Linear Algebra, Calculus & Stats are non-negotiable
OK	Code every concept — implement ML algorithms from scratch at least once
OK	Push to GitHub every day — document your experiments and results

3-MONTH OVERVIEW

M1	M2	M3
Math + Python + Data Libraries	Core ML Algorithms	Deep Learning + Deployment

MONTH 1 - Mathematical Foundations + Python Programming

Month 1	Mathematical Foundations + Python Programming
Week 1	■ Python Basics — variables, data types, control flow (if/for/while) ■ Functions, classes, modules — Python OOP fundamentals ■ Python for Data — list comprehensions, file I/O, error handling ■ Resource: Python.org Official Tutorial + Real Python ■ <i>...and more</i>
Week 2	■ Linear Algebra — vectors, matrices, dot products, eigenvalues ■ Calculus — derivatives, partial derivatives, chain rule (key for backprop) ■ Statistics & Probability — distributions, Bayes theorem, variance ■ Resource: Khan Academy + StatQuest YouTube Channel ■ <i>...and more</i>
Week 3	■ NumPy — array ops, broadcasting, vectorisation (NO for-loops in ML) ■ Pandas — DataFrames, data cleaning, groupby, merge, handling nulls ■ Matplotlib + Seaborn — visualise distributions, correlations, trends ■ Resource: NumPy Docs + Pandas Docs + Matplotlib Tutorials ■ <i>...and more</i>
Week 4	■ ML Fundamentals — supervised vs unsupervised, bias-variance tradeoff ■ Scikit-learn Basics — fit, predict, train-test split, pipelines ■ First ML Project — Iris dataset classification end to end ■ Resource: Google ML Crash Course + Kaggle Learn ■ <i>...and more</i>
BUILD	■ Data Visualisation Dashboard ■ Iris Dataset Classifier ■ ...and more mini projects

MONTH 2 - Core Machine Learning Algorithms (VERY IMPORTANT)

Month 2	Core Machine Learning Algorithms
Week 5	■ Linear Regression — cost function, gradient descent, from scratch ■ Polynomial + Ridge Regression — regularisation, overfitting ■ Logistic Regression — sigmoid, decision boundary, log-loss ■ Resource: Andrew Ng Coursera + StatQuest Logistic Regression ■ ...and more
Week 6	■ Decision Trees — Gini impurity, entropy, pruning, depth control ■ Random Forest + Ensemble Methods — bagging, boosting, voting ■ Support Vector Machines — kernel trick, hyperplane, margins ■ Resource: Visual Intro to ML + DataCamp SVM Tutorial ■ ...and more
Week 7	■ K-Means Clustering — centroid init, inertia, choosing K (elbow) ■ Hierarchical Clustering + DBSCAN — density-based, dendrograms ■ PCA — dimensionality reduction, explained variance, scree plot ■ Resource: Scikit-learn Clustering Guide + PCA Visualization ■ ...and more
Week 8	■ Cross-Validation — k-fold, stratified, leave-one-out ■ Metrics — accuracy, precision, recall, F1, ROC-AUC (know all of them) ■ Feature Engineering + Hyperparameter Tuning — GridSearchCV ■ Resource: ML Mastery Metrics + Kaggle Feature Engineering Course ■ ...and more
BUILD	■ Credit Card Fraud Detector ■ Customer Segmentation App ■ End-to-End ML Pipeline ■ ...and more

MONTH 3 - Deep Learning + Advanced Topics + Deployment

Month 3	Deep Learning + Advanced Topics
Week 9	■ Perceptron + Activation Functions — ReLU, sigmoid, tanh ■ Backpropagation — chain rule in practice, gradient flow ■ TensorFlow/Keras Intro — Sequential API, layers, compile, fit ■ Project: MNIST digit classifier — your first neural network ■ ...and more
Week 10	■ CNN Architecture — convolution, pooling, filters, feature maps ■ Popular CNNs — LeNet, AlexNet, VGG (understand, don't memorise) ■ Build + Train a CNN — image classification on CIFAR-10 ■ Resource: Stanford CS231n + Keras CNN Examples ■ ...and more
Week 11	■ RNN + LSTM Basics — sequence modeling, vanishing gradient problem ■ NLP Fundamentals — tokenization, embeddings, word2vec concept ■ Sentiment Analysis Project — train LSTM on text data ■ Resource: Understanding LSTM Networks + TensorFlow Text ■ ...and more
Week 12	■ Transfer Learning — fine-tune pre-trained models (MobileNet, ResNet) ■ GANs + Autoencoders — generative models, latent space concept ■ Deployment — save model, Flask API, host on Render/Hugging Face ■ Capstone: Complete end-to-end ML project for portfolio ■ ...and more
BUILD	■ Image Classifier Web App ■ Sentiment Analysis Model ■ Capstone Deep Learning Project ■ ...final portfolio

TARGET BREAKDOWN - What You Will Know After 3 Months

Phase	What You Will Know
Month 1	Python proficiency + strong math base + data manipulation skills
Month 2	6+ core ML algorithms implemented + model evaluation expertise
Month 3	Neural networks + CNN + RNN + deployed ML app on the internet

RESOURCES - Everything You Need to Become an AI/ML Engineer

DOCUMENTATION & PRACTICE PLATFORMS

Scikit-learn Docs	scikit-learn.org	The ML bible — every algorithm explained with code examples and math
TensorFlow / Keras	tensorflow.org/tutorials	Official tutorials for building neural networks — start here for Deep Learning
Kaggle	kaggle.com/learn	Free micro-courses + competitions + datasets — best hands-on ML platform
Syntax Error	syntaxerrrr.com	Our website — notes, one-shot videos, AI/ML + placement resources

YOUTUBE RESOURCES

STATS + ML THEORY	StatQuest with Josh Starmer by Josh Starmer youtube.com/@statquest The BEST channel for Stats and ML concepts. Every algorithm explained visually with clear statistical intuition — essential for Week 2 Math and all of Month 2 algorithms.
MATH + NEURAL NETS	3Blue1Brown by 3Blue1Brown youtube.com/@3blue1brown Visual intuition for Linear Algebra, Calculus and Neural Networks. The 'Neural Networks' series is MANDATORY watching before Week 9 backpropagation — nothing better exists.
PRACTICAL PROJECTS	Sentdex by Harrison Kinsley youtube.com/@sentdex Practical Python ML and deep learning tutorials — TensorFlow, PyTorch, real projects from scratch. Best for coding along during Month 2 and Month 3 project weeks.
AI RESEARCH + NEWS	Two Minute Papers by Karoly Zsolnai-Fehér youtube.com/@TwoMinutePapers Latest AI/ML research papers explained in simple language. Watch to stay up to date with what's happening in AI — great motivation and awareness during Month 3.

PRO TIPS FROM SYNTAX ERROR

- StatQuest is your BIBLE for understanding — watch every video on your topic
- Implement algorithms FROM SCRATCH first, then use scikit-learn — you'll understand 10x better
- Kaggle notebooks > reading theory — look at top solutions after you attempt each problem
- Math is non-negotiable — if you skip it, you WILL hit a wall at Month 3
- Consistency beats intensity — 2 hours daily for 3 months > 10 hours for 2 weeks

SYNTAX ERROR

Our Website

syntaxerrorr.com

Our Instagram

[@code.abhii07](https://www.instagram.com/code.abhii07)

"The best time to start was yesterday. The second best time is NOW. Open your notebook and write one line of code."